

LDACS1310: Introduction to Applied Cryptography

Lecture 2: symmetric encryption

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Reminder



<https://perso.uclouvain.be/fstandae/LDACCS1310/>

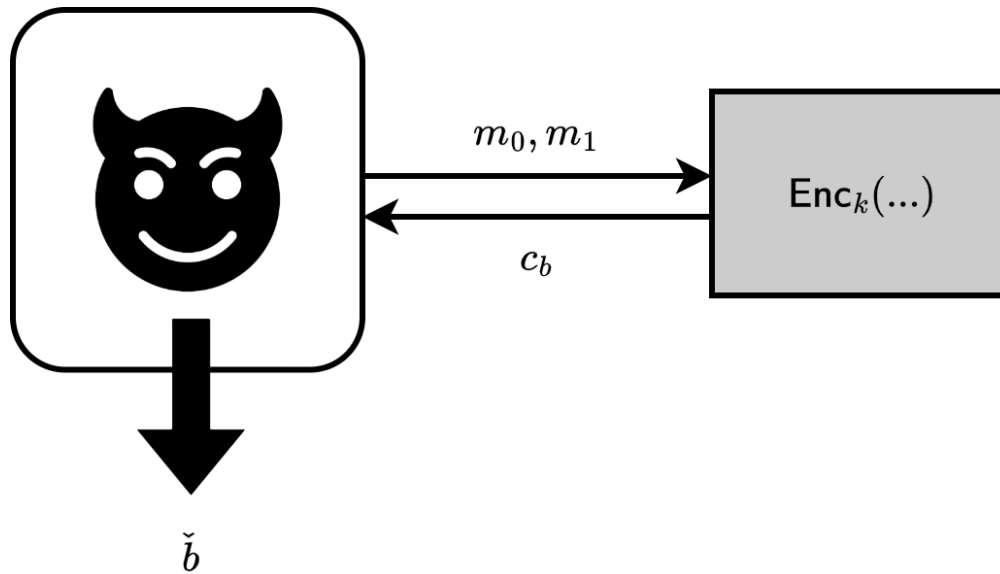
And questions from last lecture?

Assumptions dictionary

- Pseudo-random generator: $G : \{0,1\}^n \rightarrow \{0,1\}^l : k \rightarrow G(k), l > n$
$$|\Pr(\text{Dist}(G(k)) = 1) - \Pr(\text{Dist}(R) = 1)| \leq \text{negl}(n)$$
- PR function: $F : \{0,1\}^n \times \{0,1\}^m \rightarrow \{0,1\}^o : x \rightarrow F_k(x)$
$$|\Pr(\text{Dist}^{F_k(\cdot)} = 1) - \Pr(\text{Dist}^{f(\cdot)} = 1)| \leq \text{negl}(n)$$
- PR permutation: $P : \{0,1\}^n \times \{0,1\}^m \rightarrow \{0,1\}^m : x \rightarrow P_k(x)$
$$|\Pr(\text{Dist}^{P_k(\cdot)} = 1) - \Pr(\text{Dist}^{p(\cdot)} = 1)| \leq \text{negl}(n)$$

Eavesdropper security (I)

- How to **define** confidentiality in our threat model? {...}
- Eavesdropper experiment (single-message attacks) {...}



How long can the messages be?

- An encryption scheme $\text{SE} = (\text{Gen}, \text{Enc}, \text{Dec})$ is secure against eavesdropping if $\forall$ polytime Adv , $\Pr(\text{Exp}_{\text{SE}}^{\text{eav}}(n) = 1) \leq \frac{1}{2} + \text{negl}(n)$

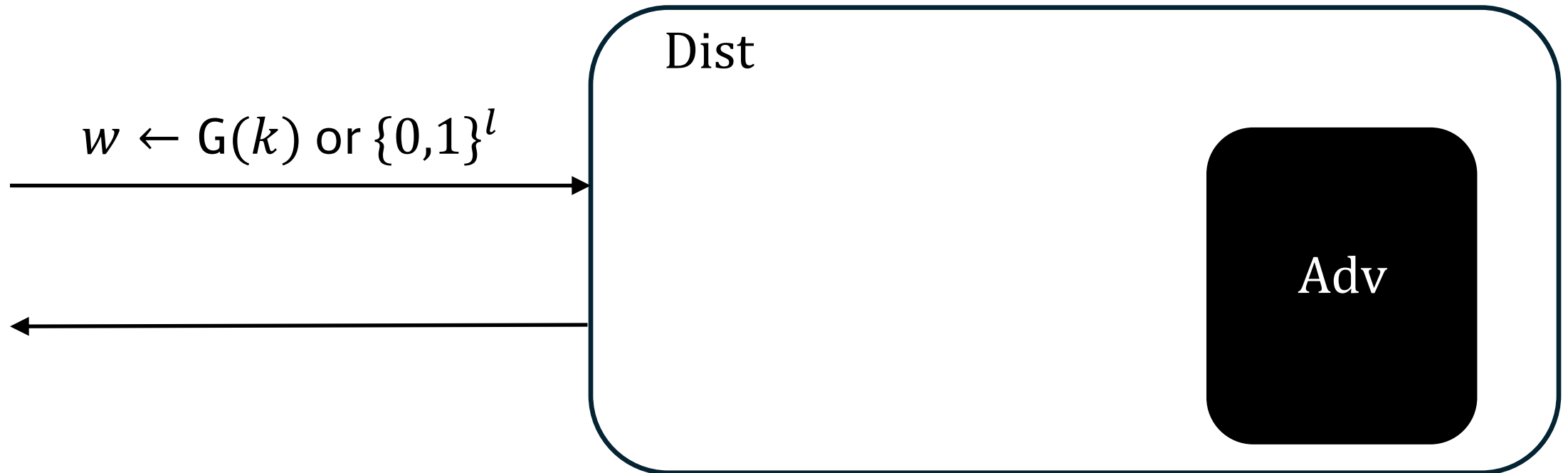
Eavesdropper security (II)

- High-level **design** idea: emulate one-time pad with a PRG
 - $c = \text{Enc}_k(m) = m \oplus G(k)$, $m = \text{Dec}_k(c) = c \oplus G(k)$

□ Question: if using the one-time pad, $\Pr(\text{Exp}_{\text{SE}}^{\text{eav}}(n) = 1) =$

Eavesdropper security (II)

- High-level **design** idea: emulate one-time pad with a PRG
 - $c = \text{Enc}_k(m) = m \oplus G(k)$, $m = \text{Dec}_k(c) = c \oplus G(k)$
- Security **reduction**: if Adv. Breaks SE, she must distinguish G



Concrete security and limitations

- If G is a (t, ϵ_G) PRG, SE is $(t - c, \epsilon_G)$ secure against eavesdropping
 - With c a constant capturing the “additional” work of Dist (vs. Adv)

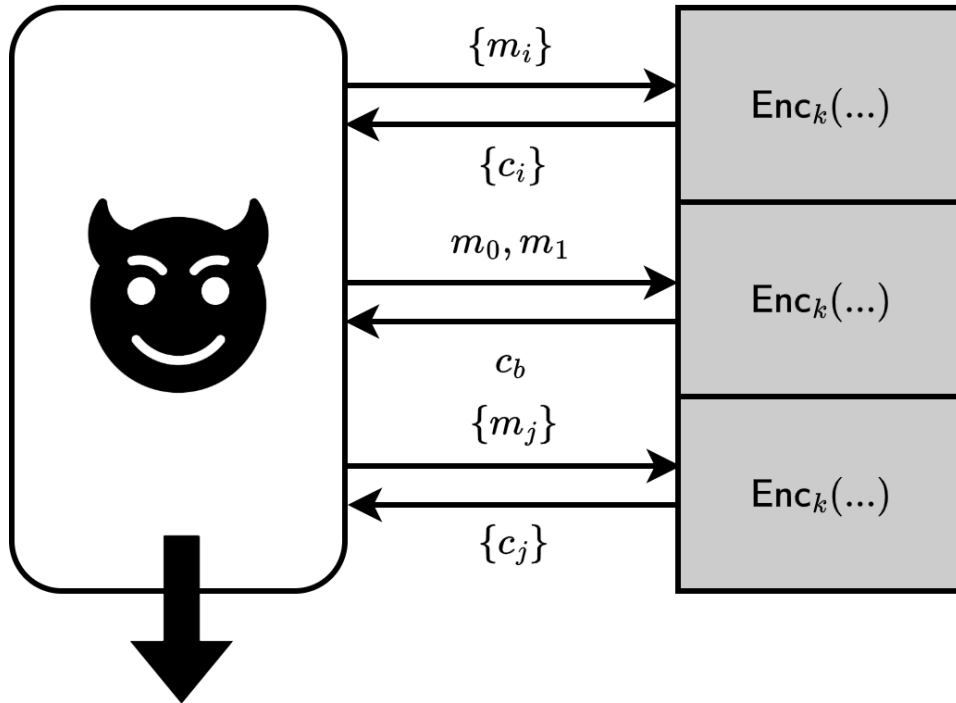
□ Question: as designers, do we hope c to be small or big?

- Limitations of eavesdropper security
 - 1.
 - 2.

⇒ We need a **stronger security definition** to cover this case

Security against Chosen Plaintext Attacks (CPA)

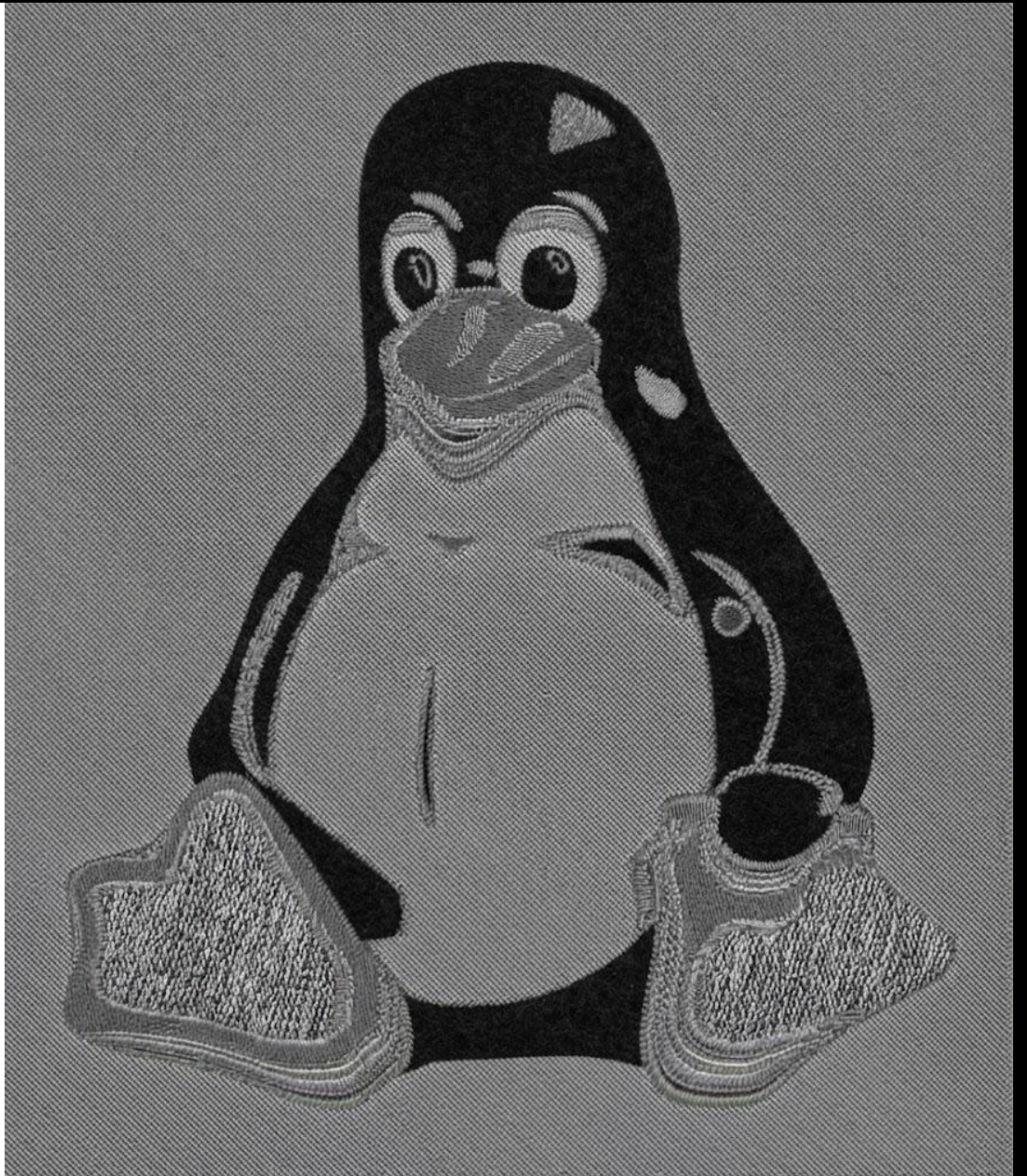
- CPA security {...}. How many queries can be made?



What if $\text{Enc}_k(m)$ always outputs the same ciphertext c ?

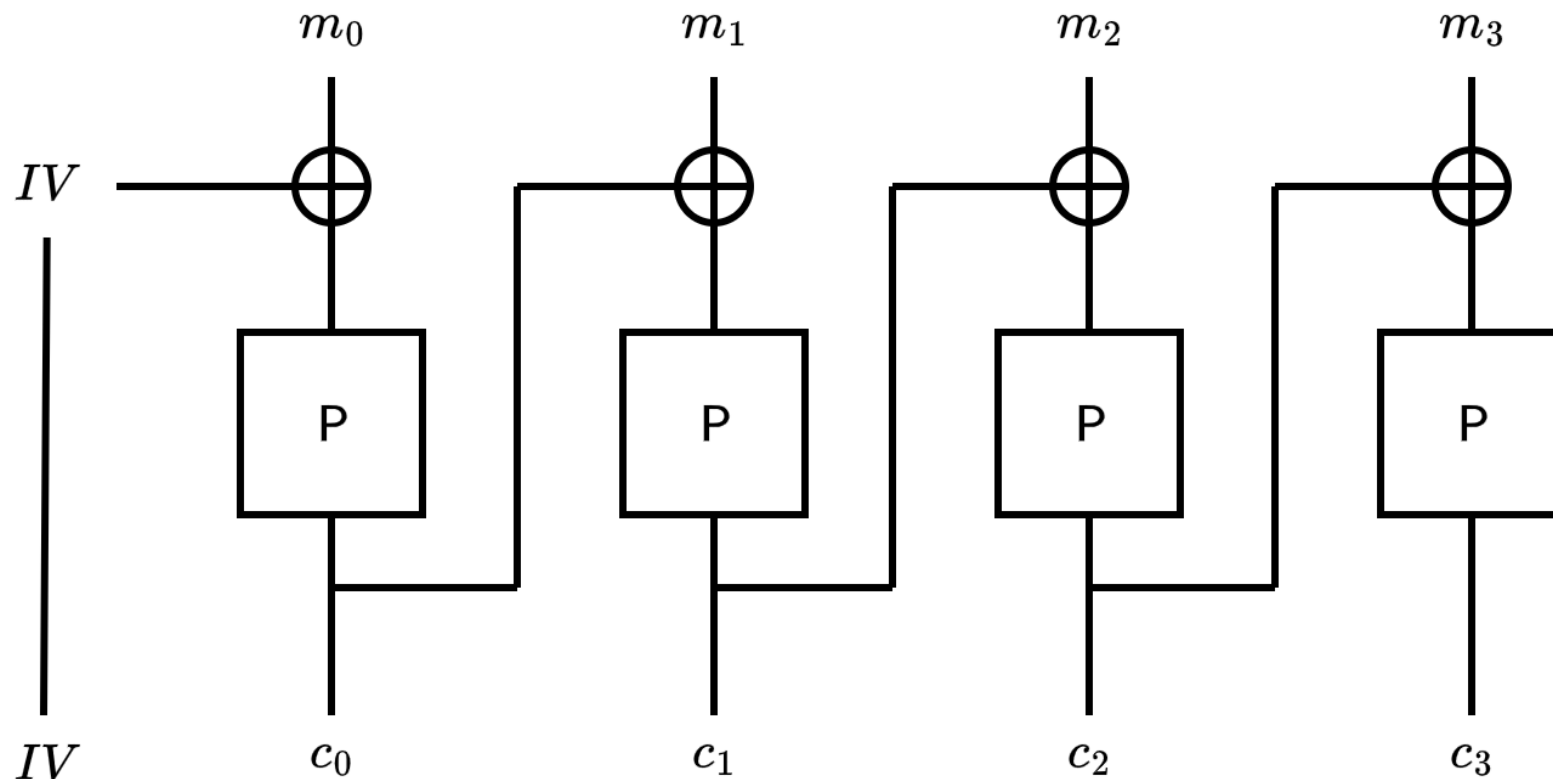
- An encryption scheme $\text{SE} = (\text{Gen}, \text{Enc}, \text{Dec})$ is secure against CPA attacks if $\forall$ polytime Adv , $\Pr(\text{Exp}_{\text{SE}}^{\text{cpa}}(n) = 1) \leq \frac{1}{2} + \text{negl}(n)$

Does it matter?



Cipher Block Chaining (CBC) encryption

- If P is a (t, ϵ_P) **PRP** with $m = n$ and IV is a random initialization vector, CBC is $(t - q \cdot c, \epsilon_{cpa})$ –secure with $\epsilon_{cpa} \leq 2 \cdot \epsilon_P + \frac{(q \cdot l)^2}{2^n}$

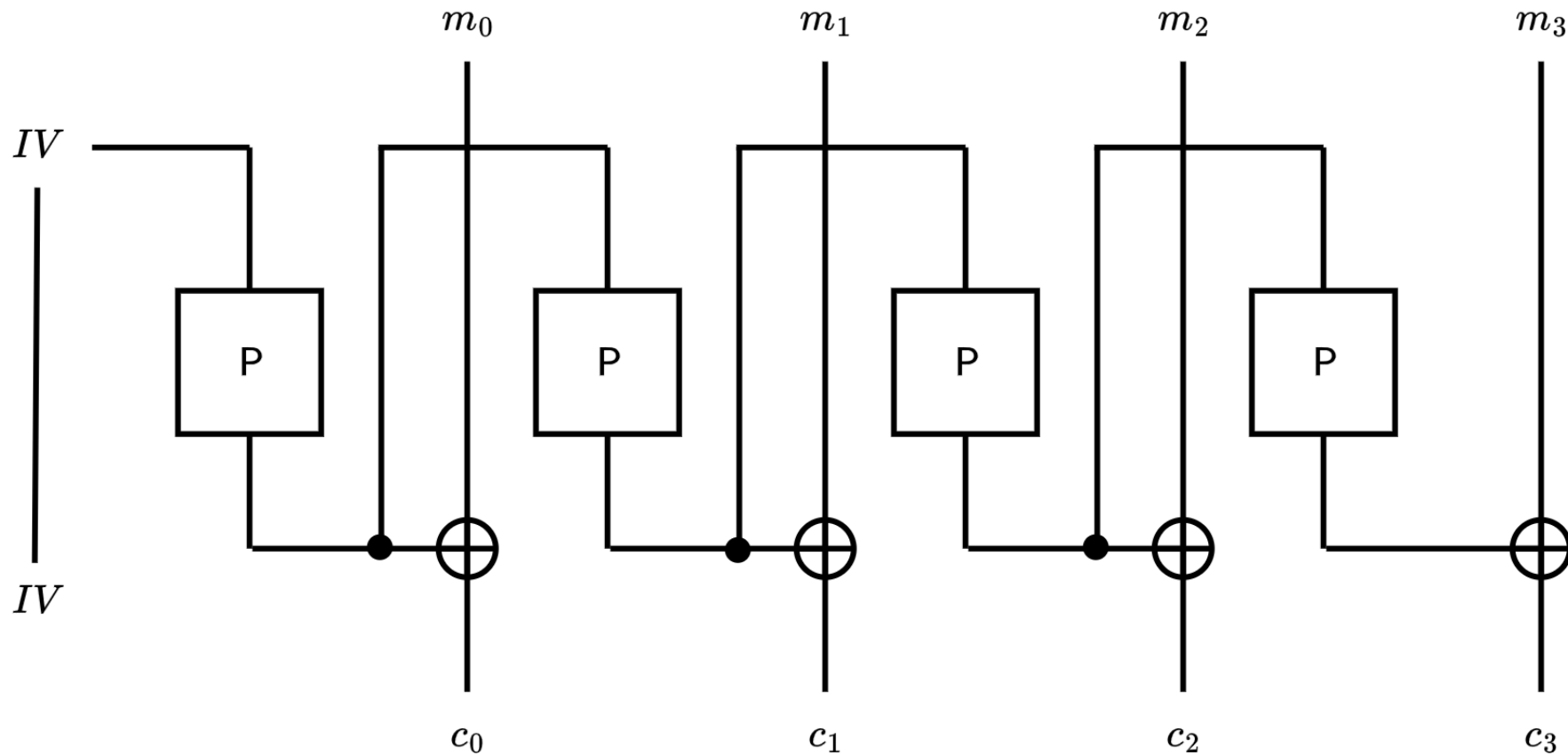


Is the bound
tight {...}?
(e.g., say $q = 1$)

What happens if
 IV is fixed? {...}

Fixed IV caveats

❑ Is it better or worse with the OFB mode below? {...}



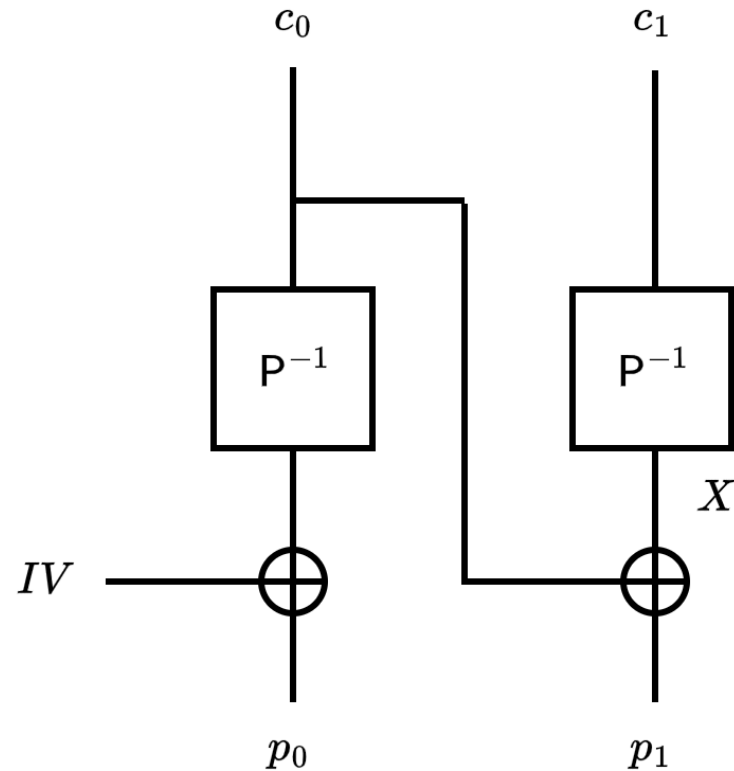
⇒ Can we weaken this requirement (see **authenticated encryption**)

Message padding (PKCS #7 standard)

- Say the permutation P operates on 16 byte, if the last block
 - Is full: add one block $16_{\text{dec}} || 16_{\text{dec}} || 16_{\text{dec}} || \dots || 16_{\text{dec}}$
 - Has one byte b_0 : pad as $b_0 || 15_{\text{dec}} || 15_{\text{dec}} || \dots || 15_{\text{dec}}$
 - Has two bytes b_0, b_1 : pad as $b_0 || b_1 || 14_{\text{dec}} || 14_{\text{dec}} || \dots || 14_{\text{dec}}$
 - Same rule up to fifteen more bytes
- Allows encryption messages of any (byte) length
- Decryption then checks that the last block is well padded

Padding oracle attacks

- Say Adv wants to decrypt c_1 and can learn whether p_1 is valid



How to proceed?

- Chosen Ciphertext Attacks (**CCA**) can be prevented with a **MAC**

Questions?